Vacation-rental marketplace — production architecture

Read-heavy browse/search, write-critical booking & payments. Services scale independently; data is partitioned by ownership.

Clients & Edge

- **Web (Next.js SSR/ISR)** — Listing pages pre-rendered + revalidated on change; islands hydrate booking widget.
- **iOS / Android** — Same public API via BFF; offline cache of trips.
- **Global CDN + WAF** — Caches HTML/ISR, JS, images (AVIF/WebP resized on the fly). Bot & DDoS protection.
- **Image service** — Upload → object storage → async resize/transcode → CDN URLs.

▼ HTTPS

API layer

- **API Gateway / BFF** — Auth (OAuth2/JWT), rate limits, request shaping per client, GraphQL or REST.
- **Service mesh** — mTLS, retries, timeouts, circuit breakers between services.

▼ gRPC / REST

Domain services (Kubernetes, autoscaled)

- **Listing** — Listings, photos, amenities, host edits.
- **Search** — Geo + date + filters, ranking, personalisation.
- **Availability & Pricing** — Calendars, dynamic price, holds with TTL.
- **Booking** — Saga: hold → pay → confirm; idempotency keys.
- **Payments** — PSP integration, payouts, ledger (double-entry).
- **Users / Reviews / Messaging** — Profiles, reviews, host–guest chat (WebSocket).

▼ events (outbox pattern) | ▼ reads/writes

Async & Search

- **Event bus (Kafka)** — listing.updated, booking.confirmed, price.changed… partitioned by listing id.
- **Search index (OpenSearch/Elasticsearch)** — Geo-sharded; near-real-time updates from CDC; availability bitmap per listing.
- **Workers** — Notifications, email/SMS, image pipeline, ranking features, fraud scoring.

Storage

- **PostgreSQL (primary + read replicas)** — Bookings, payments, users. Sharded by region/user id; strong consistency for money and inventory.
- **Redis cluster** — Sessions, hot listing cache, availability holds (TTL), rate-limit counters.
- **Object storage (S3)** — Photos, documents; lifecycle to cold tiers.
- **Data lake + warehouse** — CDC → lake → warehouse for analytics, pricing and ranking models.

Deploy & Ops

- **CI/CD** — PR checks (types, tests, visual + a11y regression) → canary → progressive rollout; feature flags.
- **Infrastructure as code** — Terraform; multi-AZ per region; active-active reads, region-pinned writes.
- **Observability** — OpenTelemetry traces, metrics, logs; SLOs on search p95 and booking success.
- **Security** — Secrets manager, PCI scope isolated to Payments, PII encryption at rest.

Frontend scaling

Listing pages are ISR-cached at the CDN, so most page views never reach origin. Images are the largest payload: resized per viewport and served from the edge.

Backend scaling

Stateless services scale horizontally on CPU/RPS. Booking uses a saga with idempotency keys so retries never double-book or double-charge.

Storage scaling

Relational store for money and inventory (correctness first), sharded as volume grows; Redis absorbs hot reads; CDC feeds search and analytics without loading the primary.

Search scaling

Index sharded by geography; queries hit only the relevant shards. Availability is pre-computed per listing so date filters don't require joins at query time.